

Area Laws in Quantum Nash Equilibria

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Abstract

We study the entanglement structure of Nash equilibria in multi-player quantum games on ring geometries, where players' strategies are local unitaries acting on a shared fiducial state represented as a matrix product state (MPS). Using simultaneous differential best response dynamics with adaptive learning rate scheduling, we numerically compute Nash equilibria for quantum Prisoner's Dilemma games with 3 and 6 players. For diagonal (commuting) payoff Hamiltonians, the welfare-optimal Nash equilibrium consistently resides on the GHZ orbit, and this robustness persists under the addition of non-commutative perturbations. We observe that low bond dimension MPS ($\chi = 2$) suffice to capture the optimal equilibria, suggesting an area-law entanglement structure for the best Nash equilibrium. We discuss the role of exploitability thresholds and the topology of the Nash variety in producing sudden welfare drops during optimization.

1 Setup

1.1 Quantum game formulation

We consider an L -player quantum game where each player i controls a single qubit. The joint state of all players is described by a quantum state $|\psi\rangle \in (\mathbb{C}^2)^{\otimes L}$. Each player's strategy consists of a single-qubit unitary $U_i \in \text{SU}(2)$ applied to their qubit. Given a strategy profile $(U_1, \dots, U_L)$ applied to a fiducial state $|\psi_0\rangle$, the resulting state is

$$|\psi\rangle = (U_1 \otimes U_2 \otimes \dots \otimes U_L) |\psi_0\rangle. \quad (1)$$

The payoff for player i is defined as the expectation value of a Hermitian operator H_i :

$$E_i = \langle \psi | H_i | \psi \rangle. \quad (2)$$

In the quantum Prisoner's Dilemma, the Hamiltonians H_i are diagonal in the computational basis,

$$H_i = \sum_{\mathbf{s} \in \{0,1\}^L} h_i(\mathbf{s}) |\mathbf{s}\rangle \langle \mathbf{s}|, \quad (3)$$

where $h_i(\mathbf{s})$ assigns a real-valued payoff to each measurement outcome $\mathbf{s} = (s_1, \dots, s_L)$.

1.2 Payoff structure

For $L = 3$ players, the payoff values are given in Table 1. The game is symmetric under cyclic permutation of the players:

$$h_i(\mathbf{s}) = h_{i+1 \bmod 3}(\sigma(\mathbf{s})), \quad (4)$$

where σ denotes the corresponding cyclic permutation of the bit string. Mutual cooperation ($\mathbf{s} = 000$) yields a payoff of 6 per player, while mutual defection ($\mathbf{s} = 111$) yields only 2—the characteristic tension of the Prisoner’s Dilemma. A single defector receives 10 while the remaining cooperators receive 3 or 0, creating an incentive to deviate from cooperation.

Outcome $\mathbf{s}$	$h_1(\mathbf{s})$	$h_2(\mathbf{s})$	$h_3(\mathbf{s})$
000	6	6	6
001	3	3	10
010	3	10	3
011	0	6	6
100	10	3	3
101	6	0	6
110	6	6	0
111	2	2	2

Table 1: Payoff values for the 3-player quantum Prisoner’s Dilemma.

For general L , players are arranged on a ring with pairwise interactions. Each player interacts with their two nearest neighbors via a 2-player Prisoner’s Dilemma Hamiltonian $H^{(2)}$ with payoffs $(R, S, T, P) = (3, 0, 5, 1)$, and the total payoff for player i is the average over incident edges:

$$H_i = \frac{1}{2} \left(H_{i,i+1}^{(2)} + H_{i,i-1}^{(2)} \right), \quad (5)$$

where indices are taken modulo L .

1.3 Matrix product state representation

For large L , storing the full state vector $|\psi\rangle \in (\mathbb{C}^2)^{\otimes L}$ is intractable. We represent quantum states as matrix product states (MPS) with open boundary conditions:

$$|\psi\rangle = \sum_{s_1, \dots, s_L} A_1^{s_1} A_2^{s_2} \cdots A_L^{s_L} |s_1 s_2 \cdots s_L\rangle, \quad (6)$$

where each $A_j^{s_j}$ is a matrix of dimension $\chi_{j-1} \times \chi_j$ with $\chi_0 = \chi_L = 1$. The bond dimension $\chi = \max_j \chi_j$ controls the entanglement capacity of the ansatz. For $\chi = 1$, the state is a product state (zero entanglement); for $\chi = 2^{\lfloor L/2 \rfloor}$, the MPS can represent any state exactly.

In practice, we work with bond dimension $\chi = 2$, which is sufficient to capture states with moderate entanglement, including GHZ-type states $|\text{GHZ}\rangle = (|0\rangle^{\otimes L} + |1\rangle^{\otimes L})/\sqrt{2}$.

1.4 Nash equilibrium and exploitability

A state $|\psi\rangle$ constitutes an ε -Nash equilibrium if no player can increase their payoff by more than ε through a unilateral unitary deviation:

$$\max_{U_i \in \text{SU}(2)} \langle \psi_i(U_i) | H_i | \psi_i(U_i) \rangle - \langle \psi | H_i | \psi \rangle < \varepsilon \quad \forall i, \quad (7)$$

where $|\psi_i(U_i)\rangle$ denotes the state with U_i applied only to qubit i . The left-hand side defines the *exploitability* $\varepsilon_i(\psi)$ of player i .

We find Nash equilibria using *simultaneous differential best response dynamics*. At each iteration n , we compute the gradient of each player's payoff with respect to their local unitary:

$$M_i = \frac{\partial E_i}{\partial U_i} = \text{tr}_{j \neq i} [H_i |\psi\rangle\langle\psi|], \quad (8)$$

where the partial trace is over all qubits except i . The unitary update is extracted via the SVD of the matrix $\mathbb{I} - \alpha M_i$: if $\mathbb{I} - \alpha M_i = Y \Sigma Z^\dagger$, the update is $U_i = ZY^\dagger$, which is the unitary closest to the gradient step in operator norm. All players update simultaneously, and the procedure is iterated until the total exploitability $\sum_i \varepsilon_i < \varepsilon_{\text{thr}}$.

The exploitability ε_i is computed by globally optimizing over $\text{SU}(2)$ using differential evolution, ensuring that the reported Nash equilibria are validated against arbitrary (not just infinitesimal) deviations.

1.5 Algorithmic refinements

Several algorithmic improvements were necessary to obtain reliable Nash equilibria across a range of system sizes and Hamiltonian types.

Subroutine learning rate scheduling. When Nash equilibria are not found by the differential best response solver, the failure is typically due to insufficient learning rates or premature termination, rather than the algorithm converging to a non-Nash state. We address this by implementing adaptive retry logic in the solver: when convergence fails, the subroutine learning rate α is increased and the solver is restarted. Analysis of failed runs confirms this diagnosis:

- At low learning rates ($\alpha = 0.01$), the exploitability decreases but does not reach the threshold within the allotted iterations—a premature termination artifact.
- At moderate learning rates ($\alpha \approx 0.09$), convergence improves significantly.
- If α exceeds a critical threshold ($\alpha \gtrsim 3$), the exploitability exhibits limit cycles and fails to settle, an artifact of overshooting in the best response dynamics.
- An intermediate regime ($\alpha \approx 2$) can produce states with constant payoff but nonzero exploitability—locally stable but globally unstable configurations.

Note that α does not directly set the step size; it scales the matrix $\mathbb{I} - \alpha M_i$ whose SVD yields the unitary update, so the effective step size also depends on the gradient magnitude. These regimes are illustrated in Figure 1.

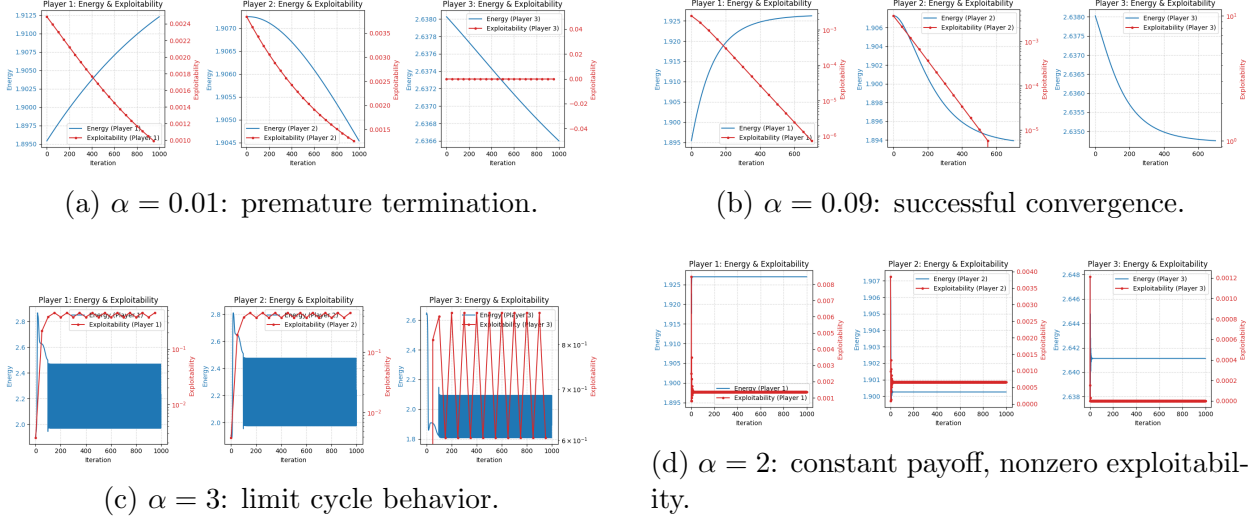


Figure 1: Effect of the subroutine learning rate α on the convergence of the differential best response solver for 3 players.

Fiducial state learning rate and exploitability threshold. The outer optimization loop over the fiducial state $|\psi_0\rangle$ is sensitive to the exploitability threshold ε_{thr} used in the inner Nash solver. At $\varepsilon_{\text{thr}} = 10^{-4}$, the solver may converge to different Nash equilibria on the same orbit at successive optimization steps, producing sudden welfare drops (see Section 2.3). Tightening to $\varepsilon_{\text{thr}} = 10^{-10}$ eliminates these discontinuities but reduces the success rate of the inner solver, as some runs fail to converge entirely. The combination of tighter thresholds with the adaptive learning rate scheduling resolves this trade-off.

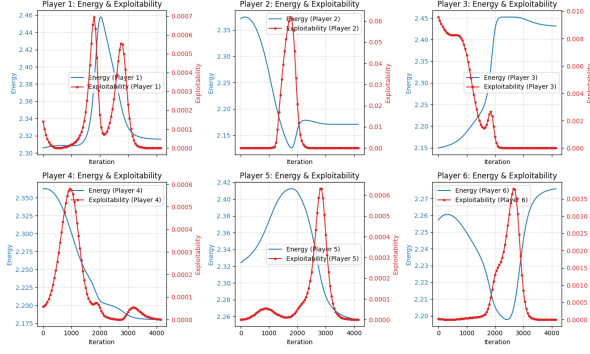
Multiple Nash equilibria per orbit. When the solver fails, an alternative strategy is to perturb the state within the same orbit (by applying products of random local unitaries) and re-run the algorithm. This keeps optimization running but reveals that multiple Nash equilibria can coexist on the same orbit, manifested by jumping payoffs across restarts. For the 6-player case, the landscape is particularly complex (Figure 2), with the solver requiring up to 20 random restarts to find a valid equilibrium.

2 Numerical results: diagonal payoff Hamiltonians

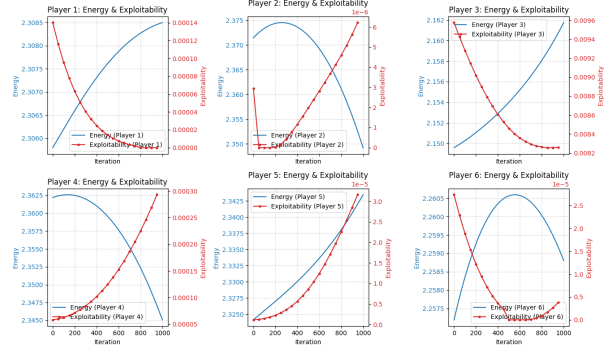
2.1 Three-player quantum Prisoner's Dilemma

For the 3-player game with the payoff structure in Table 1, the welfare-optimal Nash equilibrium is found on the GHZ orbit, $|\psi_0\rangle \sim (|000\rangle + |111\rangle)/\sqrt{2}$. The optimization of the fiducial state consistently converges to this orbit across multiple random initializations.

Not all runs succeed: Figure 3 shows a case where the algorithm converges to a locally stable state (at iteration 8401) that has low exploitability by the local criterion but remains globally exploitable.



(a) $\alpha = 0.4$, 6000 iterations: convergence achieved.



(b) $\alpha = 0.06$, 1000 iterations: not converged.

Figure 2: Solver behavior for the 6-player game at different learning rates. The more complex landscape requires higher learning rates and more iterations.

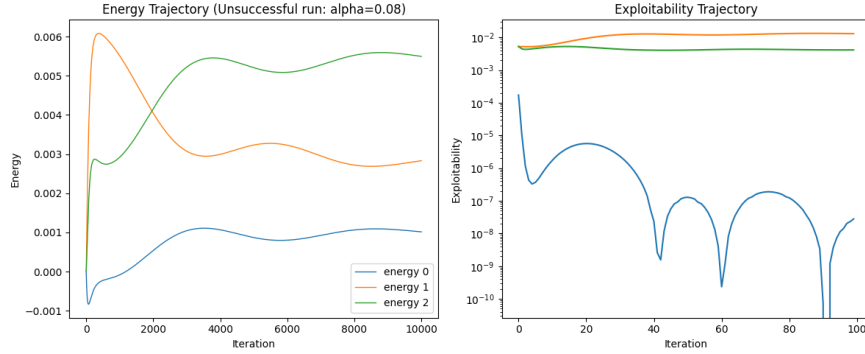


Figure 3: An unsuccessful 3-player run: the solver reports convergence to a Nash state, but the global exploitability remains high, indicating a locally stable but globally unstable strategy configuration.

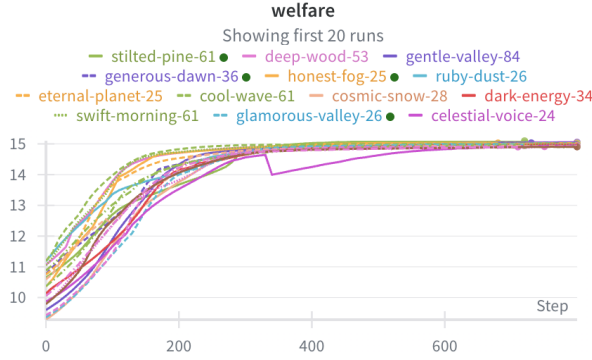
2.2 Six-player results and bond dimension scaling

For $L = 6$ players with the ring-geometry pairwise Hamiltonian, we swept over bond dimensions $\chi \in \{2, 3, 4, 5, 6, 7, 8, 9\}$ with randomly initialized MPS states. Counter-intuitively, higher bond dimensions do not uniformly improve outcomes: the best welfare is observed at low χ , while larger χ runs exhibit more frequent welfare drops due to the richer Nash landscape (Figure 4).

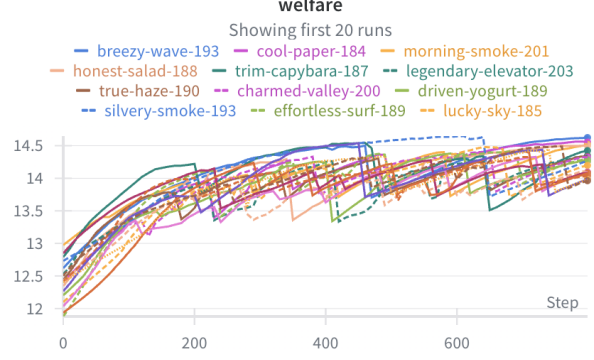
The MPS with bond dimension $\chi = 4$ already captures the full state space for $L \leq 5$ qubits without truncation, as the number of MPS parameters exceeds the true degrees of freedom of the Hilbert space. The GHZ state, describable with $\chi = 2$, remains attainable for all truncations under consideration.

2.3 Welfare drops and the topology of the Nash variety

A persistent phenomenon in the fiducial state optimization is the occurrence of sudden welfare drops. These drops are robust against:



(a) $\chi = 2$: smooth welfare optimization.

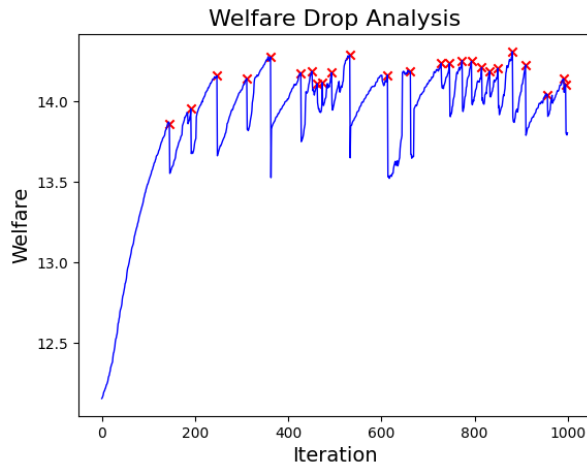


(b) $\chi = 8$: frequent welfare drops.

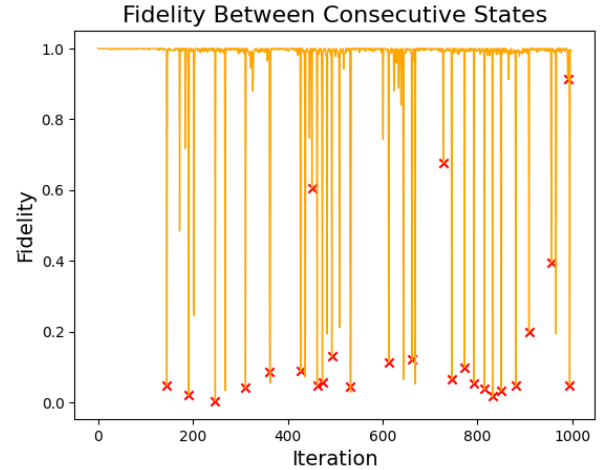
Figure 4: Welfare trajectories for 6-player optimization at different bond dimensions. Lower bond dimensions yield smoother optimization landscapes.

1. Changing the gradient calculation to select the maximum-welfare perturbation.
2. Adding ridge regularization to the gradient estimator.

The welfare drops arise because the manifold of Nash equilibria in the pure state space does not have consistent topology across different regions (Figure 5). At an ε -approximate Nash equilibrium with $\varepsilon = 10^{-4}$, a small perturbation of the fiducial state can cause the inner solver to converge to a qualitatively different Nash equilibrium with drastically different payoffs.



(a) Welfare trajectory showing sudden drops.



(b) Corresponding payoff trajectories.

Figure 5: Case study of sudden welfare drops during fiducial state optimization. The drops are robust against changes to the gradient estimator and regularization.

The entanglement parameters (bipartite entropies), by contrast, change smoothly across these drops (Figure 6). This is expected: the orbit is updated continuously, so the entangle-

ment structure varies smoothly even when the Nash equilibrium found on that orbit changes discontinuously.

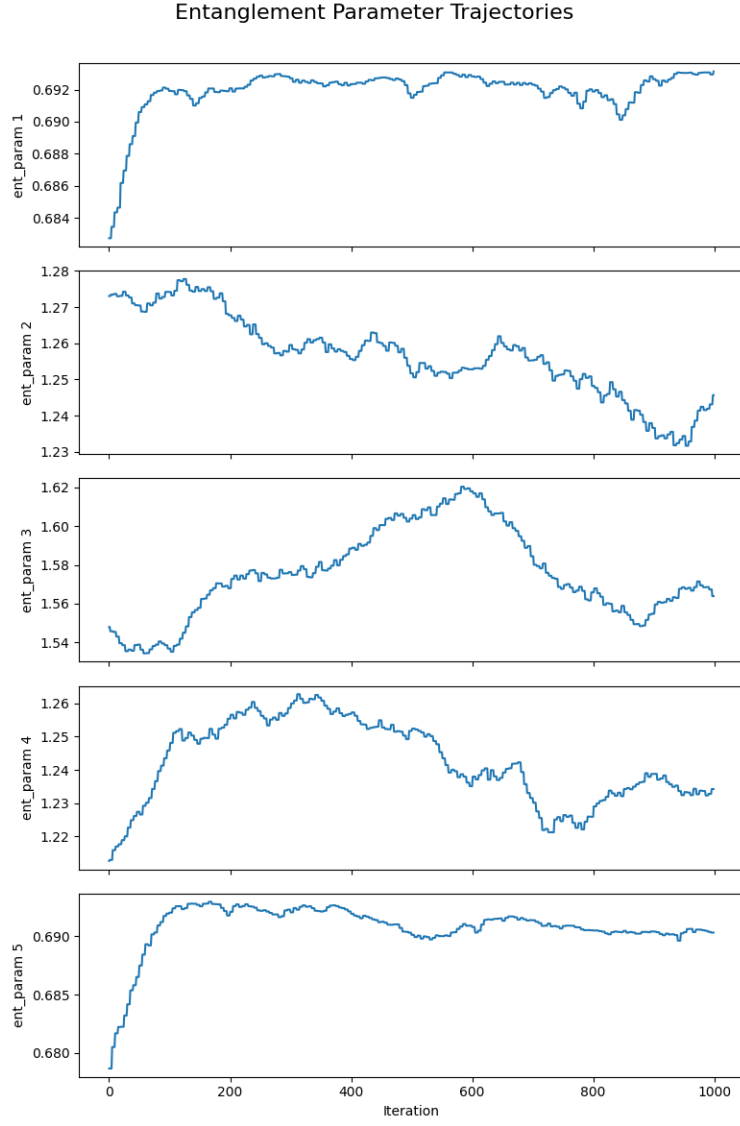


Figure 6: Entanglement parameters evolve smoothly during optimization, even across welfare drops, confirming that the orbit changes continuously.

Gradient diagnostic analysis reveals:

- The gradient norm increases sharply at larger perturbation strengths ϵ , indicating that large gradients are artifacts of basin-jumping between Nash equilibria.
- The success rate of the inner solver drops significantly at larger ϵ .
- The normalized gradient (welfare change divided by ϵ) varies wildly, confirming that the gradient estimate becomes unreliable when perturbations cross basin boundaries.

These diagnostics are summarized in Figure 7. The resolution is to reduce the perturbation strength ϵ such that 2ϵ is smaller than the “bottleneck” of the connected component in the Nash variety, ensuring that the optimization remains on a single continuous branch.

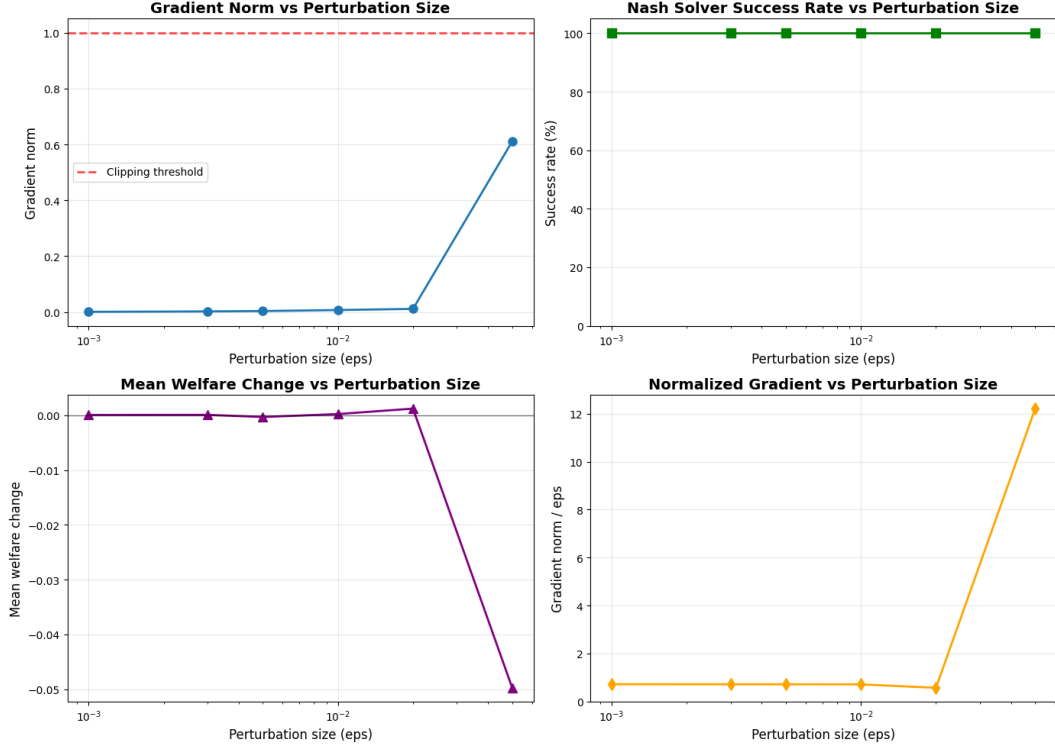


Figure 7: Gradient diagnostics for the welfare drop analysis. Top left: gradient norm vs. perturbation strength. Top right: solver success rate. Bottom left: mean welfare change. Bottom right: normalized gradient. The sharp increase in gradient norm and drop in success rate at larger ϵ confirm basin-jumping as the cause of welfare drops.

3 Area laws and entanglement structure

A quantum game is defined by the triple $(\hat{h}_i, G_i, N)$ where N is the number of players, $\hat{h}_i$ are the payoff Hamiltonians, and G_i are the strategy groups. We study the quantum advantage associated with the entanglement structure of the underlying resource state.

- In the product state orbit ($\chi = 1$), the game is purely classical, with automatic extension to mixed strategies by controlling the probability amplitudes of individual qubits.
- In the volume-law regime, local unitary invariance implies $\text{tr}(U_i \rho U_i^\dagger \hat{O}) \approx \text{tr}(\rho \hat{O})$ for any local observable $\hat{O}$, so players have negligible strategic influence.
- There may therefore exist an optimal entanglement structure for the fiducial state that maximizes the quantum advantage in the sense of mechanism design—neither too little nor too much entanglement.

Our numerical results support this picture: the welfare-optimal Nash equilibria for the quantum Prisoner’s Dilemma consistently reside on the GHZ orbit ($\chi = 2$), and higher bond dimensions do not improve the optimal welfare. This suggests that the best Nash equilibrium satisfies an area law for entanglement, with the optimal resource state having $O(1)$ entanglement entropy across any bipartition.

4 Non-commutative payoff Hamiltonians

4.1 Introducing non-commutativity

To move beyond the diagonal (classical) payoff structure, we consider Hamiltonians with non-commuting terms. The key observations, organized by the degree of non-commutativity, are:

Product state regime. For product states, Nash equilibria are always found irrespective of the payoff Hamiltonians. The algorithm converges reliably in this regime.

Ising-type perturbations. Adding entanglement via a multi-qubit unitary, one can find Nash equilibria for random Ising games with

$$\hat{h}_L = \hat{h}_0 + V(g_L), \quad \hat{h}_R = \hat{h}_0 + V(g_R), \quad (9)$$

where $V(g) = g \hat{X}_i$ and g_L, g_R are drawn from a random distribution.

Heisenberg-type interactions. The existence of Nash equilibria breaks down for random Heisenberg interactions

$$\hat{h}_i = n_x \hat{X}_i \hat{X}_{i+1} + n_y \hat{Y}_i \hat{Y}_{i+1} + n_z \hat{Z}_i \hat{Z}_{i+1}, \quad (10)$$

where (n_x, n_y, n_z) is a random unit vector on S^2 . Adding entanglement in the initial states causes the best response dynamics to end up in limit cycles, preventing convergence to a Nash equilibrium.

Remark. All games considered so far are translationally invariant, i.e., $T^\dagger \hat{h}_i T = \hat{h}_i$ where T is the translation operator.

4.2 Three-player results with non-commutative perturbations

To introduce non-commutativity gradually, we let $\hat{h}' = \hat{h} + V$ where V is a random Hermitian matrix with bounded Frobenius norm. For $\|V\|_F = 1$, the fiducial state optimization still converges to the GHZ orbit (Figure 8), suggesting robustness of the GHZ orbit as the welfare-optimal state even in the presence of non-commuting perturbations.

The relationship between entanglement parameters and equilibrium welfare remains qualitatively the same across different non-commutativity strengths (Figure 9). The success rate

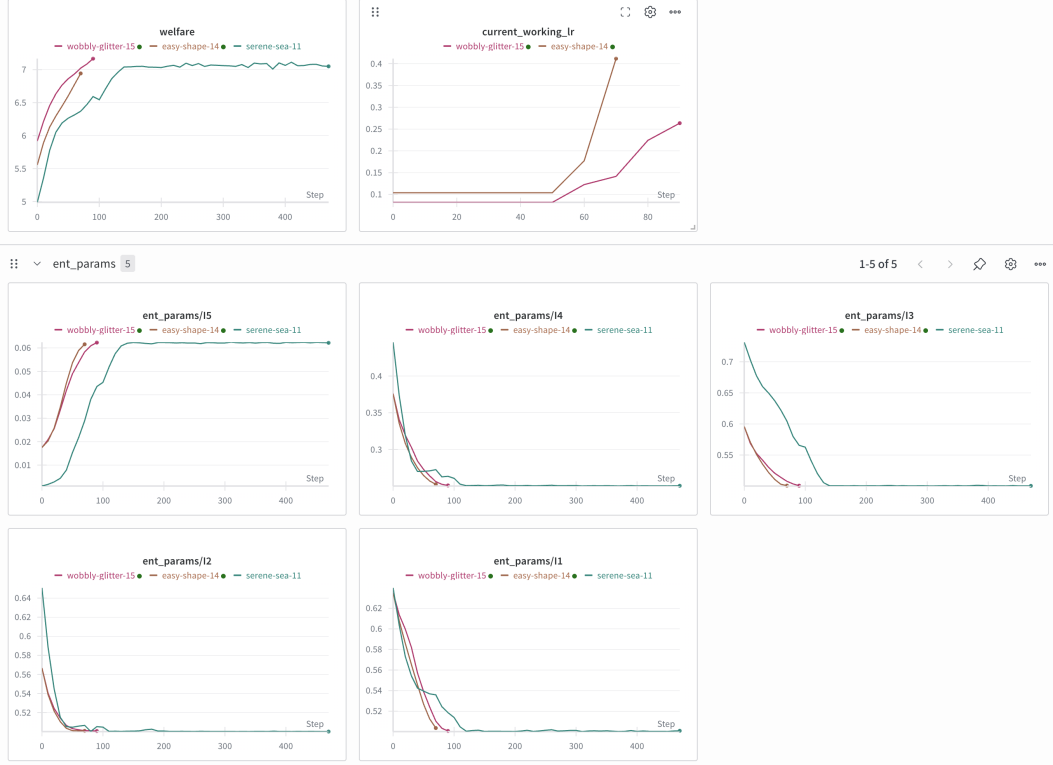


Figure 8: Fiducial state optimization with non-commutative perturbations ($\|V\|_F = 1$). Compared to the standard (diagonal) QPD runs, the optimization still converges to the GHZ orbit, suggesting robustness of maximal entanglement as the optimal resource state.

of the inner Nash solver drops to approximately 60%, but the optimal orbit remains the GHZ orbit when equilibria are found.

The `compute_exploitability()` function was modified to handle the expanded strategy space arising from non-commuting Hamiltonians, where the single-player best response is no longer one-dimensional.

4.3 Two-player non-commutative games

For 2-player games, we map out the exploitability landscape as a function of the entanglement in the fiducial state (Figure 10):

- Local solvers fail to find Nash equilibria once the state has nonzero Schmidt eigenvalues (low-lying entangled orbits).
- Convergence to Nash equilibria is successful in the product state regime.
- Using damped-step best response dynamics destabilizes the optimization, while full-step best response gives correct results.

Explicit search over the full strategy space confirms that the exploitability does not reach zero for entangled states with non-commutative payoffs, with the minimal exploitability consistent with the local algorithm at approximately 0.01. This suggests a genuine obstruction

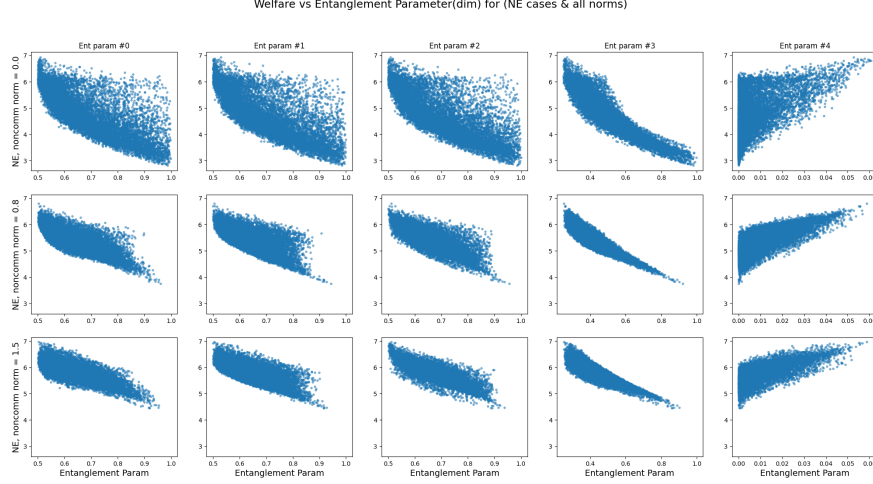


Figure 9: Entanglement parameters vs. welfare for Nash equilibria found under different non-commutativity strengths. The relationship is qualitatively invariant.

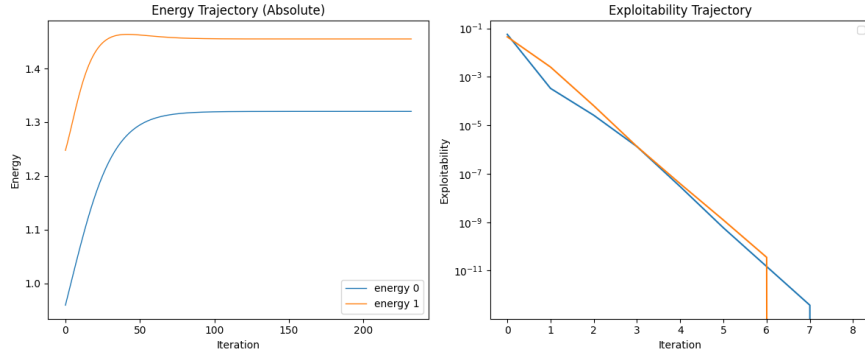


Figure 10: Two-player non-commutative game with product states. The algorithm converges successfully in the product state regime, but fails for entangled states with nonzero Schmidt eigenvalues.

to Nash equilibria in the entangled regime for non-commuting Hamiltonians, rather than a failure of the algorithm.

4.4 Open questions

1. Sequential (instead of simultaneous) updates for the best response solver may improve convergence for non-commutative payoffs.
2. A differential best response solver without SVD decomposition could avoid the instabilities observed with damped steps.
3. Double-oracle methods for expanding subgames may provide a path to equilibria in cases where the current algorithm fails.
4. CMA-ES navigation on the exploitability landscape could find minima that local gra-

dient methods miss, though the computational cost (requiring differential evolution subroutine calls for each update) may be prohibitive.

5. Testing with completely random payoff Hamiltonians: does the optimal orbit remain the maximally entangled one?